HOTEL BOOKING AND CANCELLATION ANALYSIS

Project Overview and Objective

This project involves cleaning, transforming, and analysing a real-world hotel booking dataset of approximately 119K records using Python (Pandas, NumPy, Matplotlib and Seaborn) in Google Collab, in order to understand customer booking and cancellation behaviour in the hospitality and tourism domain.

The main objectives are:

- To identify meaningful patterns, trends, and insights from the dataset using exploratory data analysis (EDA).
- To clean, transform, and preprocess the data for analysis.
- To create meaningful visualizations that clearly explain the story behind the data.
- To provide actionable insights and recommendations from the analysis.

Expected outcome: The project uncovers key patterns, trends and hidden insights from the dataset and supports decision-making with evidence-based findings.

Data Sources

- **Source Description and Timeline:** Open hotel booking dataset (csv format) sourced from GitHub; data collected during 2017–2019.
- **Dataset Size:** 119,390 rows and 29 columns (86,129 rows and 36 columns after cleaning and feature engineering).
- **Domain:** Hospitality and Tourism.

Problem Statement

- To determine which hotel type (City Hotel or Resort Hotel) is more popular and how booking volume differs between them.
- To analyse the impact of stay duration on cancellation rates.
- To study the relationship between booking lead time and cancellations.
- To identify monthly booking trends and seasonal patterns, including peak and low-demand periods.
- To develop business recommendations for improving revenue, reducing cancellations, optimizing seasonal operations and managing advance bookings.

Attribute (Column / Features) Details

Attribute Name	Data Type	Description
hotel	Categorical	Type of hotel booked: City Hotel or Resort Hotel

Attribute Name	Data Type	Description
is_canceled	Binary (0/1)	Indicates whether the booking was cancelled
lead_time	Numeric (Integer)	Number of days between booking and arrival
arrival_date_year	Numeric (Integer)	Year of the guest's arrival
arrival_date_month	Categorical	Month of the guest's arrival
arrival_date_week_number	Numeric (Integer)	Week number of the arrival date
arrival_date_day_of_month	Numeric (Integer)	Day of the month of arrival
stays_in_weekend_nights	Numeric (Integer)	Number of weekend nights stayed
stays_in_weekdays_nights	Numeric (Integer)	Number of weekday nights stayed
adults	Numeric (Integer)	Number of adults in the booking
children	Numeric (Float)	Number of children in the booking
babies	Numeric (Integer)	Number of babies in the booking
meal	Categorical	Meal plan selected by the guest
city	Categorical	Guest's city/location information
market_segment	Categorical	Market segment through which the booking was made
distribution_channel	Categorical	Channel used to distribute or receive the booking
is_repeated_guest	Binary (0/1)	Indicates whether the guest is a returning customer
previous_cancellations	Numeric (Integer)	Number of previous bookings cancelled by the guest
previous_bookings_not_canceled	Numeric (Integer)	Number of previous bookings not cancelled
booking_changes	Numeric (Integer)	Number of changes made to the booking
deposit_type	Categorical	Type of deposit required for the booking
agent	Categorical / ID	ID of the travel agent associated with the booking
company	Categorical / ID	ID of the company associated with the booking
days_in_waiting_list	Numeric (Integer)	Number of days the booking stayed on the waiting list

Attribute Name	Data Type	Description
customer_type	Categorical	Type/category of customer making the booking
adr	Numeric (Float)	Average Daily Rate charged for the room
required_car_parking_spaces	Numeric (Integer)	Number of parking spaces requested
total_of_special_requests	Numeric (Integer)	Total number of special requests made by the guest
reservation_status	Categorical	Final status of the reservation (Check-Out, Canceled, No-Show)

Tools & Technologies

- **Google Colab:** Environment for writing and running the analysis.
- **Python – Pandas & NumPy:** Data loading, cleaning, transformation and feature engineering.
- **Matplotlib & Seaborn:** Creation of all charts and visualizations (bar, line, donut, histogram, box plot, heatmap).
- **Power BI:** Dashboard creation.

Data Pre-Processing (Python / Pandas)

Tasks Performed:

- **Data Loading & Inspection:** Loaded hotel_bookings_data.csv with pandas (skipping bad lines) and inspected shape (119,390 × 29), column names, data types and summary statistics.
- **Missing Value Treatment:** Found nulls in children (4) and city (488), plus agent and company. Filled children with 0 (assumed no children), city with "Unknown", and agent/company with "No Agent" / "No Company".
- **Duplicate Removal:** Detected 33,261 duplicate records and removed them to maintain data accuracy; the duplicate count was verified as 0 afterwards.
- **Data Validation:** Re-checked null values and data types before visualization. The cleaned dataset contains 86,129 records.

Feature Engineering (Python)

Seven new features were derived to support deeper analysis, taking the dataset from 29 to 36 columns:

New Feature	Data Type	Logic / Description
total_stay_nights	Numeric	Weekend nights + weekday nights
total_guests	Numeric	Adults + children + babies
guest_nights	Numeric	Total guests × total stay nights
guest_type	Categorical	Single (1 guest), Couple (2 guests), Group (3+ guests)
stay_category	Categorical	Short (0–2), Medium (3–5), Long (6–10), Very Long (10+ nights)
lead_time_category	Categorical	Last Minute (0–7), Short Notice (8–30), Medium Advance (31–90), Long Advance (91–180), Very Long Advance (180+ days)
guest_status	Categorical	Returning Guest or New Guest, from is_repeated_guest

---

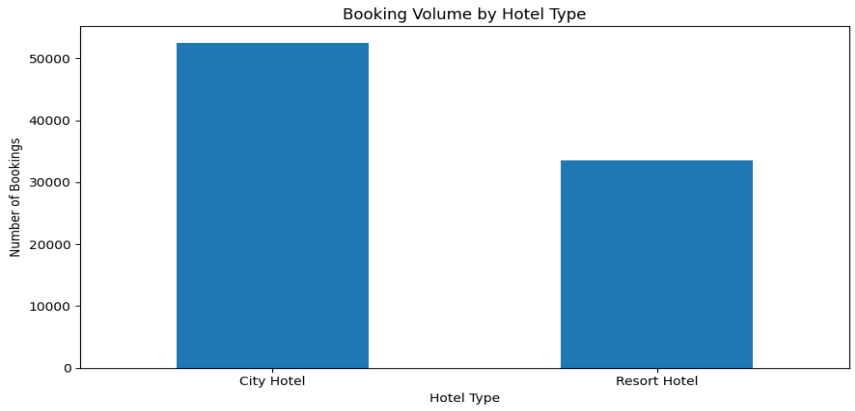
Analysis and Visualizations (Python)

EDA approach:

- **Univariate analysis:** Hotel type, market segment and customer type (new vs repeated).
- **Bivariate analysis:** Hotel type vs cancellation, hotel type vs ADR, stay duration vs cancellation, lead time vs cancellation, and month vs hotel type.
- **Multivariate analysis:** Correlation heatmap of lead time, special requests, cancellation and other booking variables.

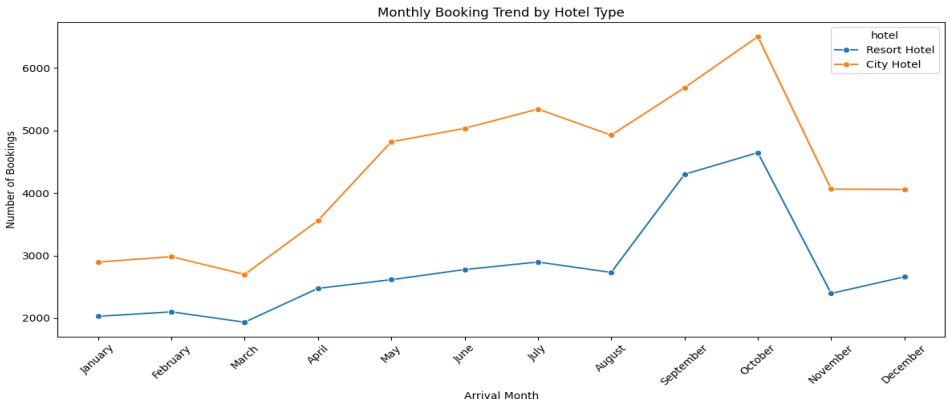
All ten charts produced in the notebook are included below, each with its key insight.

Chart 1: Booking Volume by Hotel Type *(Univariate – Bar chart)*



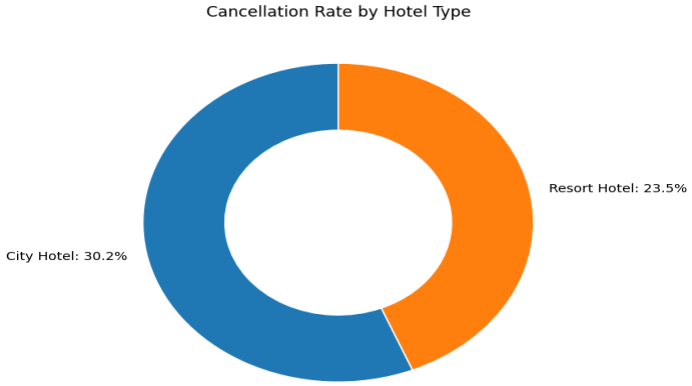
Insight: City Hotel has about 52,500 bookings and Resort Hotel about 33,500, so City Hotel accounts for roughly 61% of total booking volume.

Chart 2: Monthly Booking Trend by Hotel Type *(Bivariate – Line chart)*



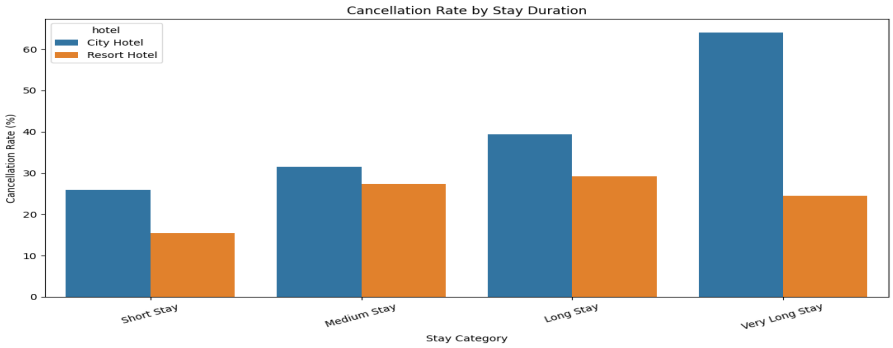
Insight: Bookings are low in January–March (City about 2,700–3,000; Resort about 1,900–2,100). They climb from April and peak in September–October (City about 6,500 and Resort about 4,650 in October), then drop in November–December. Resort jumps more sharply from August (about 2,700) to September (about 4,300).

Chart 3: Cancellation Rate by Hotel Type *(Bivariate – Donut chart)*



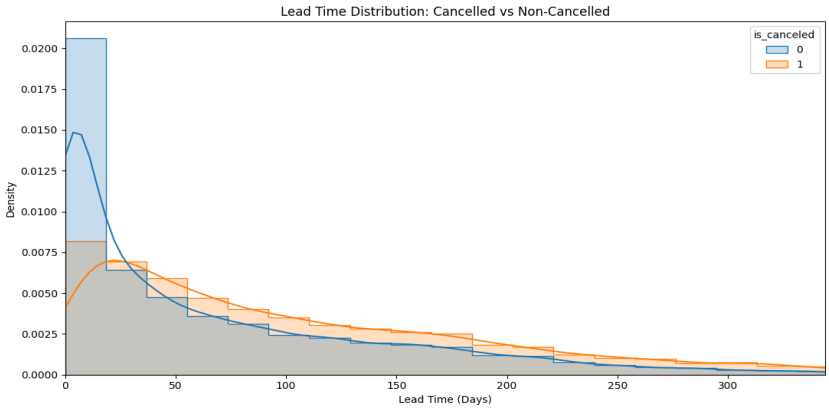
Insight: City Hotel cancels about 30.2% of bookings and Resort Hotel about 23.5%, a gap of roughly 7 percentage points. Overall, the cancellation rate is about 28%, so more than 1 in 4 bookings is lost.

Chart 4: Cancellation Rate by Stay Duration *(Bivariate – Grouped bar chart)*



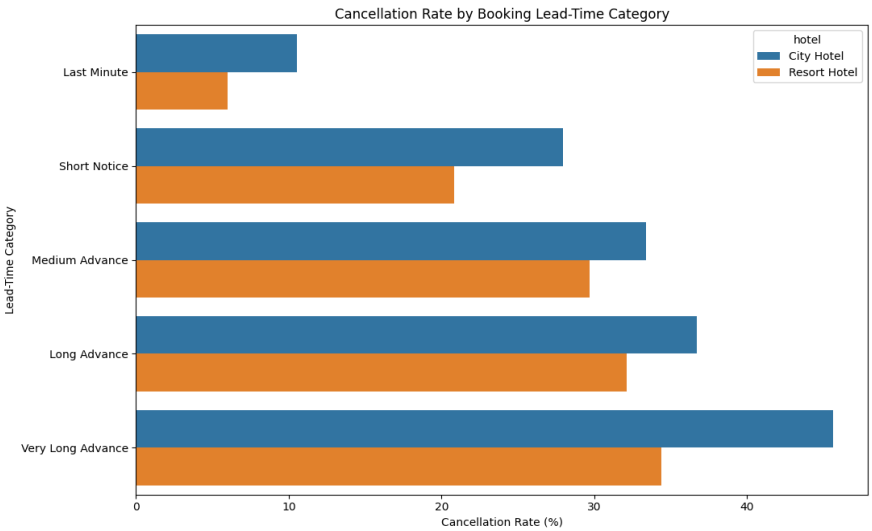
Insight: City Hotel's cancellation rate rises with stay length: about 26%, 31.5%, 39% and then 64% for very long stays. Resort Hotel stays fairly flat at about 15–29%, dipping to about 24.5% for very long stays.

Chart 5: Lead Time Distribution – Cancelled vs Non-Cancelled *(Bivariate – Histogram with KDE)*



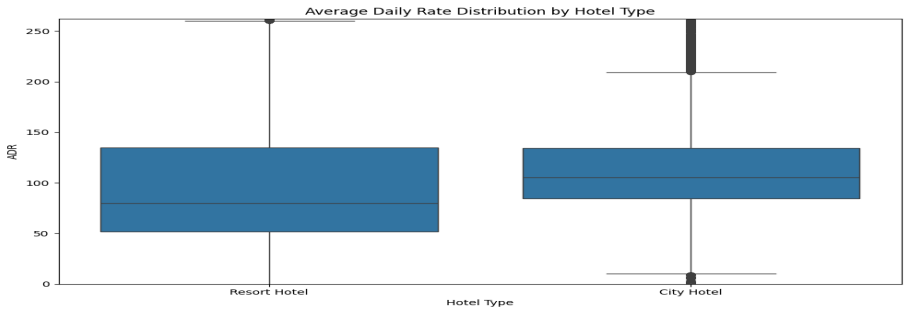
Insight: Non-cancelled bookings spike at under about 20 days of lead time, while cancelled bookings have a heavier tail beyond about 25 days. The earlier a guest books, the more likely the booking is cancelled.

Chart 6: Cancellation Rate by Booking Lead-Time Category *(Bivariate – Horizontal bar chart*



Insight: City Hotel goes from about 10.5% (0–7 days) through 28%, 33.5% and 37% to about 46% (180+ days). Resort Hotel goes from about 6% through 21%, 30% and 32% to 34%.

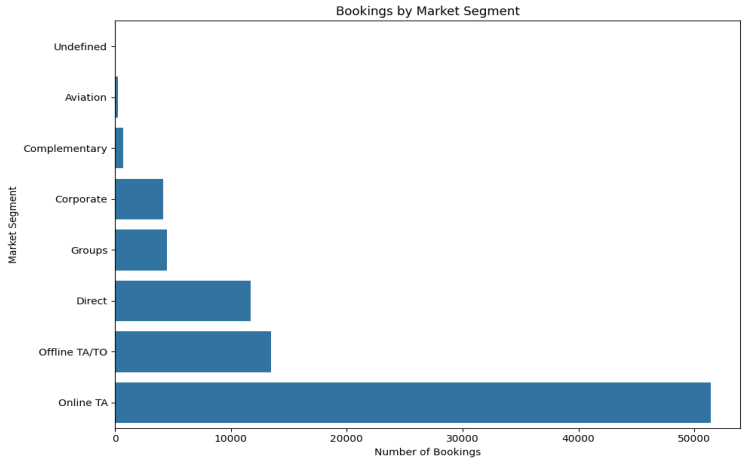
Chart 7: Average Daily Rate (ADR) Distribution by Hotel Type *(Bivariate – Box plot)*



Insight: Resort Hotel's median ADR is about 80 with a wide spread (roughly 52–135), which fits seasonal pricing. City Hotel's median is about 105 with a tighter spread and many high

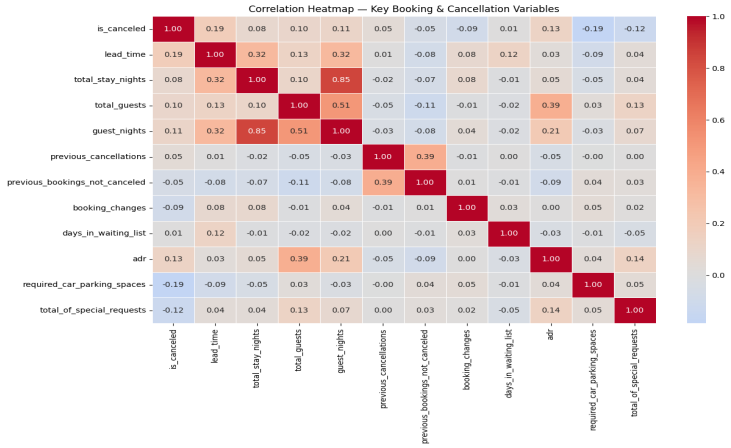
outliers. Near-zero values are likely free or complimentary bookings. The chart is clipped at the 99th percentile.

Chart 8: Bookings by Market Segment *(Univariate – Horizontal bar chart)*



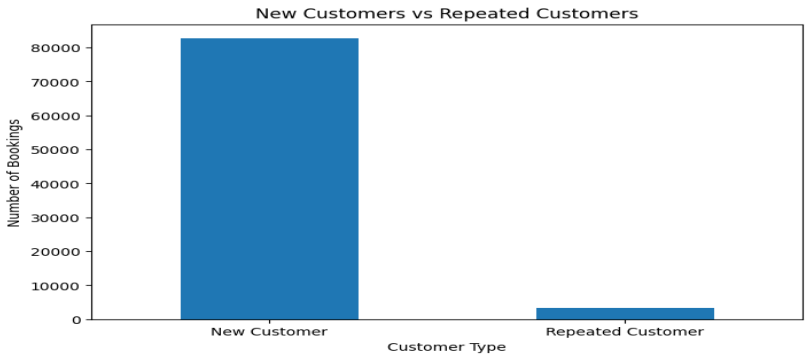
Insight: Online travel agents (OTAs) generate about 51,500 bookings, roughly 60% of the total. Offline agents and tour operators contribute about 13,500 and Direct about 11,700, while Groups and Corporate contribute about 4,500 and 4,200. The business depends heavily on OTAs.

Chart 9: Correlation Heatmap – Key Booking and Cancellation Variables *(Multivariate – Heatmap)*



Insight: The strongest links to cancellation are lead time (+0.19), parking requested (−0.19), ADR (+0.13), special requests (−0.12) and guest-nights (+0.11). All are weak (under 0.2), so cancellation has many causes and no single variable predicts it. Stay nights and guest-nights are 0.85 correlated because one was built from the other, so a model should not use both. The heatmap shows correlation, not causation, and only linear relationships.

Chart 10: New Customers vs Repeated Customers *(Univariate – Bar chart)*



Insight: The vast majority of bookings come from new customers (roughly 82,000+), while repeated customers account for only a small share (roughly 3,500, about 4%). This indicates low guest loyalty and room to grow repeat business.

Insights & Conclusions

Key Findings:

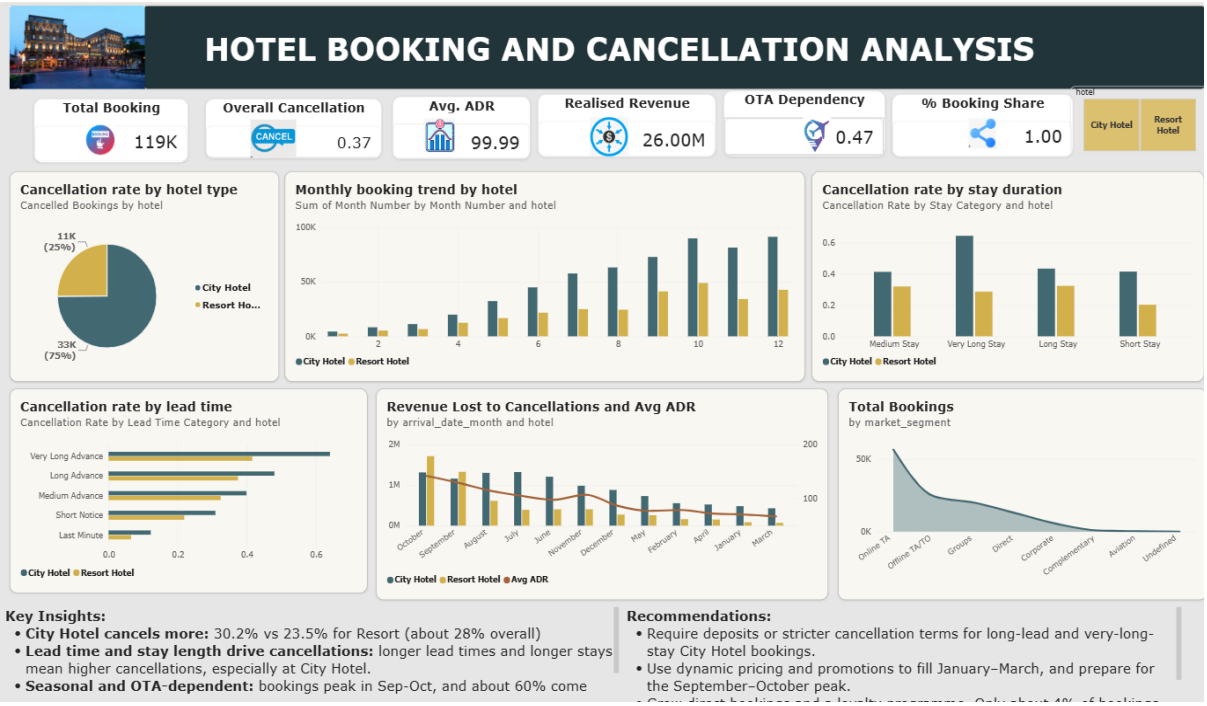
- City Hotel is the more popular property (about 61% of bookings) but also cancels more (about 30.2% vs 23.5% for Resort).
- Overall, about 28% of bookings are cancelled, so more than 1 in 4 is lost.
- Cancellation rises with lead time and, for City Hotel, with stay length (up to about 64% for very long stays).
- Demand is seasonal: low in January–March, peaking in September–October.
- About 60% of bookings come through online travel agents, and only a small share of bookings is from repeated guests.

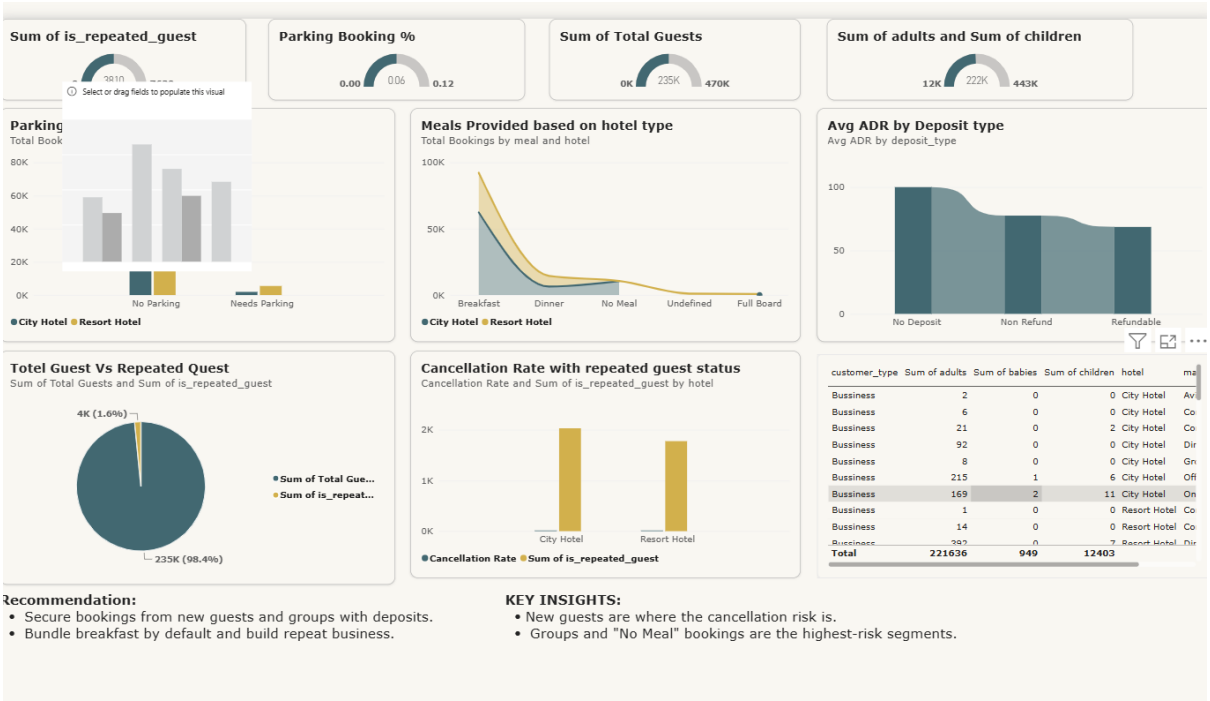
Analysis Insights:

- **Descriptive:** City Hotel has about 52,500 bookings vs about 33,500 for Resort. Median ADR is about 105 for City and about 80 for Resort. OTAs dominate the market segment mix.
- **Diagnostic:** Cancellations are driven by long lead times and long stays (especially at City Hotel). Correlations with cancellation are weak (all under 0.2), so cancellation has many causes: lead time (+0.19), parking requested (−0.19), ADR (+0.13) and special requests (−0.12) are the strongest.
- **Predictive:** Bookings made 180+ days ahead carry the highest cancellation risk (about 46% City, 34% Resort), while last-minute bookings carry the lowest (about 10.5% City, 6% Resort). Seasonal demand is expected to repeat, with peaks in September–October. Lead time and booking features can be used to build a cancellation-prediction model, avoiding using stay nights and guest-nights together because they are 0.85 correlated.

- **Prescriptive:** Introduce deposits or stricter cancellation policies for long-lead and very-long-stay City Hotel bookings; use dynamic pricing and promotions to fill January–March and capitalise on September–October peaks; manage overbooking and inventory by lead-time risk; reduce dependence on OTAs by growing direct bookings; and launch loyalty programmes to increase repeat guests.

Dashboard using Power BI





Conclusions

The Python-based EDA workflow proved effective for end-to-end analysis, from raw hotel booking data to cleaned, feature-engineered data and visual insights. The analysis shows that cancellations are a significant revenue risk (about 28% overall), concentrated in long-lead and long-stay bookings, particularly at City Hotel. Combined with strong seasonality and heavy OTA dependence, these findings give the hotel group clear, evidence-based levers for reducing cancellations, optimizing seasonal operations and improving revenue.